C2 guide: research and specification

Name _____ Section _____ Date _____

Component 2, Assessment Criterion 2. Your brief says what you will make and for whom. This section finds out what the product has to be, and ends with a specification that you can test at the end of the project.

What Cambridge says

These are the examiners' own words, from the Cambridge marking guidance for IGCSE Design and Technology (0445).

- *"Research into the design brief resulting in a specification."*
- *"Candidates should carry out appropriate research in order to develop their understanding of the identified opportunity which will lead to a justified specification."*
- Research *"could include: the analysis of existing products, the location or situation for the product, surveys/questionnaires, items to be held/stored with the main dimensions (if appropriate), anthropometric data (if appropriate), consideration of materials, hardware (such as hinges, handles, fastenings), construction methods and surface finishes."*
- *"Further information should be collected that directly relates to the user and the user's needs for the product."*
- *"Candidates should only provide information that is relevant to their product, as often page after page of generic material is seen that has no relevance to the situation being investigated."*
- *"Work presented in the research section should be fully analysed and conclusions made by explaining what has been found out and what information will be taken forward."*
- *"The specification ... should be presented as a detailed and justified description of what is required from the end product. A specification that is only a list of key words will fail to gain credit at level 2 or level 3."*

What that means for you. Do not copy pages from the internet. Collect only what your product needs, say what each piece **tells you**, and let those conclusions become the specification.

What you hand in

Tick each line when the page is in your folder.

- ☐ **A. Existing products.** One or two products like yours, analysed with ACCESS FM, with the real measurements you took.
- ☐ **B. The place.** Photographs and measurements of where the product will live.
- ☐ **C. The user.** Your interview or survey answers about what they need. If your product holds things, list **the items with their main dimensions**. If your product is used by a body, record the **body measurements** that matter (anthropometric data).

- ☐ **D. Materials, hardware and finishes.** The options that fit your product, including hardware such as hinges, handles and fastenings, construction methods and surface finishes. For each, what you learned.
- ☐ **E. Conclusions.** One page: **what I found out** on the left, **what I will take forward** on the right.
- ☐ **F. The specification.** Six or more points. Each one has a **criterion**, a **measure** and a **justification** that points back to a finding in A to E.

How to do it, in order

1. **Before you research, write the questions.** What do you need to know before you can decide the size, the material, the switch, the finish?
2. **Collect.** Measure real things. Photograph the real place. Ask your user. Take notes in your own words.
3. **Analyse as you go.** After each piece of research, write two lines: what it tells you, and what you will use.
4. **Write the conclusions page (E)** before you write the specification. If a conclusion does not change the product, delete it.
5. **Write the specification (F)** from the conclusions. Then give it to a partner and ask them to design the **worst possible product that still passes every point**. Fix every loophole they find.

The mistakes Cambridge names

Mistake	How to avoid it
Pages of generic material with no relevance to your product	Before you add a page, ask: which specification point will this change?
Information presented with no analysis or summary of findings	Every section ends with "what this tells me" and "what I will take forward".
A specification that is only a list of key words	Write each point as a full sentence: criterion, measure, reason.

A worked example (invented, same product as the C1 guide)

Marta's bedside shelf. The research and the points below are made up so you can see the standard.

What I found out	What I will take forward
The gap beside the bed is 350 mm wide and 300 mm deep (measured with a tape, photographed).	The product must be narrower than the gap, with room to slide it in.
Her items: phone 160 × 75 × 9 mm, glasses case 160 × 60 × 40 mm, paperback 200 × 130 × 25 mm.	The shelf must be at least 200 mm deep to hold the paperback flat.
A shop-bought shelf I measured has no front lip, and Marta says her phone slid off one like it.	The product needs a front lip.
She cannot see in the dark, so she finds things by touch.	The switch must be reachable by touch with one hand.

Criterion	Measure	Justification
The product must not be wider than 340 mm.	Steel rule across the widest part.	The gap beside her bed is 350 mm wide and the product must slide in.
The shelf must be at least 200 mm deep.	Steel rule on the shelf surface.	Her paperback is 200 mm long and must lie flat.
The shelf must have a front lip at least 10 mm high.	Steel rule against the lip.	Her phone slid off the lipless shelf I measured.
The switch must be on the front edge, within 100 mm of the right-hand side.	Measure from the edge.	She uses it with her right hand, by touch, in the dark.

Check before you hand it in

- Does every page of research change a specification point? If not, take it out.
- Does the conclusions page say what you will **take forward**, not only what you found?
- Does every specification point have a measure you could carry out with the equipment in the workshop?
- Could a partner find a terrible product that passes your points? If yes, tighten them.